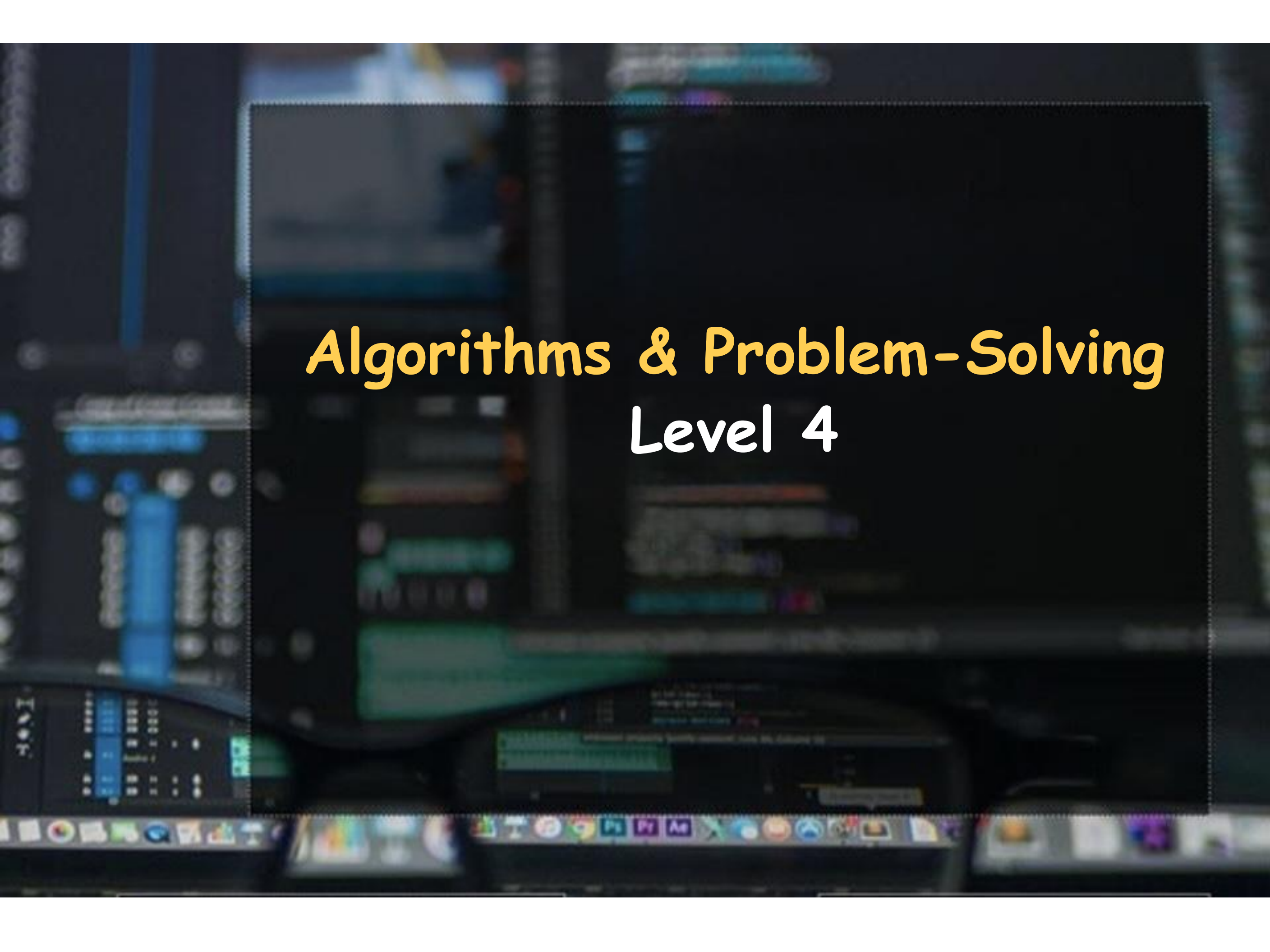




Algorithms & Problem-Solving Level 4



Algorithms & Problem Solving

Level 4

Leap Year

Definition

- The goal of this program is to check whether an input year is a leap year or not, using only one line of code within the check function.
- This is achieved by using logical operators instead of multiple if/else conditional statements.



Explanation

- The basic rule for determining a leap year is the same as in the previous problem:
 - A year is a leap year if it:
 - is divisible by 4
 - is not divisible by 100
 - is divisible by 400
 - The difference in this exercise is that all the previous rules are condensed into a single logical condition using the AND and OR operators.

Procedures

① Read the year from the user:

- Using the `readYear()` function, which prompts the user to enter the year and returns it as an integer:

```
int readYear()
{
    int year;
    cout << "Please enter a year to check? ";
    cin >> year;
    return year;
}
```

② Checking if a year is a leap year (in just one line):

- In the `isLeapYear()` function, only one logical expression is used:
`return(year % 4 == 0 && year % 100 != 0) || (year % 400 == 0);`

- Condition explanation:

`(year % 4 == 0 && year % 100 != 0)`

The year is divisible by 4 but not by 100.

`(year % 400 == 0)` Or the year is divisible by 400 (exception for centuries). The final result is true if either condition is met.

3 Displaying the result:

■ In main(), the function is called and the result is printed:

```
if (isLeapYear(year))
```

```
    cout << "\nYes, Year [" << year << "] is a leap year.\n";
```

```
else
```

```
    cout << "\nNo, Year [" << year << "] is NOT a leap year.\n";
```



Example:

Year	Result
1996	Leap Year
1900	Not Leap Year
2000	Leap Year
2004	Leap Year
2010	Not Leap Year



Note:

- The following logical operators were used:
 - && (AND) to join two conditions that must both be true
 - || (OR) to join two conditions where only one condition is true for the result to be true

Explanation

- This solution is more concise and efficient because the compiler will evaluate the logical condition in a single line without the need for multiple consecutive if statements.
- Despite its brevity, it achieves the same logic as the previous problem with complete accuracy.

Summary

- ✓ The logic of a leap year has been simplified to a single logical condition using `&&` and
- ✓ The condition is only true when:
 - The year is divisible by 4 but not by 100, or by 400

Thank you 😊